MONASH University

MONASH BUSINESS SCHOOL

ETC4500/ETC5450 Advanced R programming

Week 10: Metaprogramming

Outline

1 Metaprogramming

2 (Non-)standard evaluation

3 Tidy evaluation

Metaprogramming

Metaprogramming is *programming* about programming.

In other words, writing code that can *inspect* and *modify* code.

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A powerful idea

Unlike most programming languages, R *embraces* metaprogramming and non-standard evaluation (NSE).
This powers much of the strange but wonderful interface designs in R and its packages.

The rlang package

```
library(rlang)
```

A package for writing R code that interacts with R code.

! Not a new idea!

Metaprogramming/NSE doesn't require the rlang package.

There are base R equivalents to the functions shown today.

NSE is widely used in base R (not just in the tidyverse!)

Parsing code

Every time you run code anywhere in R it needs to be 'interpreted' by the parser.

The parser reads unstructured text (your written code) and interprets it as an expression.

```
# parse(text = "seq(1, 10, by = 0.5)")
parse_expr("seq(1, 10, by = 0.5)")
```

```
seq(1, 10, by = 0.5)
```

Deparsing code

Deparsing takes an expression and converts it back to text.

```
my_seq <- parse_expr("seq(1, 10, by = 0.5)")
expr_text(my_seq)
```

```
[1] "seq(1, 10, by = 0.5)"
```

This can be useful for providing informative error messages, or print output for objects which store expressions.

Code is data

Expressions (code) can be used like any other data in R.

```r
my_seq <- parse_expr("seq(1, 10, by = 0.5)")
my_seq
```

```
seq(1, 10, by = 0.5)
```

```r
class(my_seq)
```

```
[1] "call"
```

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my_seq
```

```
seq(1, 10, by = 0.5)
```

```r
class(my_seq)
```

```
[1] "call"
```

```r
eval(my_seq)
```

```
 [1]  1.0  1.5  2.0  2.5  3.0  3.5  4.0  4.5  5.0  5.5  6.0  6.5  7.0  7.5
[15]  8.0  8.5  9.0  9.5 10.0
```

Inspecting code

R expressions behave exactly like lists

```r
as.list(my_seq)
```

```
[[1]]
seq

[[2]]
[1] 1

[[3]]
[1] 10

$by
[1] 0.5
```

Inspecting code

They can also be subsetted to inspect the functions and arguments.

```
my_seq[[1]]
```

```
seq
```

```
my_seq[["by"]]
```

```
[1] 0.5
```

Modifying code

Expressions can be modified by replacing their elements.

```
my_seq[["by"]] <- 1
my_seq
```

```
seq(1, 10, by = 1)
```

```
eval(my_seq)
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

Looking at code

Your turn!

How do infix operators (like `+`, `*`, and `%in%`) get interpreted by the parser?

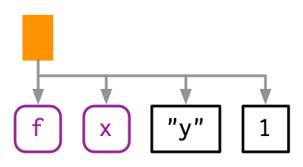
Try to parse `5 + 3 * 7`, and see how the order of operations are represented in the parsed expression.

Bonus: rewrite this expression without infix operators.

Abstract syntax trees

The structure of expressions is commonly known as an abstract syntax tree (AST). We can use `lobstr::ast()` to explore it.

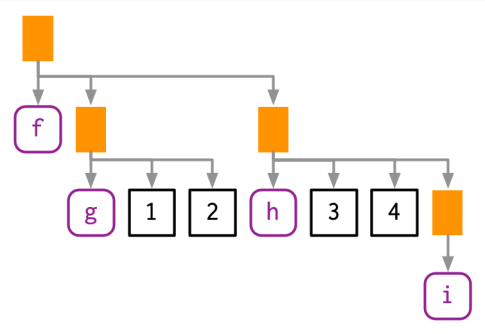
```
lobstr::ast(f(x, "y", 1))
```



Abstract syntax trees

More complicated (nested) code results in a larger/deeper AST.

```
lobstr::ast(f(g(1, 2), h(3, 4, i())))
```



Abstract syntax trees

Your turn!

Inspect the AST for the following code:

- `5 + 3 * 7`
- `mtcars |> select(cyl)`
- `mtcars |> mutate(wt/hp)`

How does R structure these expressions?

Bonus: does `-2^2` yield 4 or -4? Why?

Analysing code

How would you programmatically analyse code from hundreds of packages?

- Regular expressions on the source code? Maybe…
- Traverse the parsed source code's AST? Yes!

This however can be tricky, requiring recursive algorithms that explore the AST using breadth/depth first search (BFS/DFS).

Coding code

You can also write code that creates code. For this we use the `call2()` function

```
# call("seq", 1, 10, by = 0.5)
call2("seq", 1, 10, by = 0.5)
```

```
seq(1, 10, by = 0.5)
```

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```

```
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```

! parse_expr() or call2()?

You might be tempted to `parse()` code that you `paste()` together, but this is unsafe and unreliable! Why?

Coding code

After the break…

Recall in week 8 (shiny) I shared this function:

```
react <- function(e) new_function(alist(), expr(eval(!!enexpr(e))))
```

Next we'll learn how it uses NSE to change how code runs.

Code evaluation

Standard evaluation

- The code and environment is unchanged.
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Non-standard evaluation (NSE)

- The code and/or the environment is changed.
- Leading to the evaluated result changing.

Standard or non-standard evaluation?

Your turn!

Do the following functions use standard evaluation or NSE?

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- `mtcars |> select(cyl)`
- `read_csv("data/study.csv")`
- `ggplot() + geom_line()`
- `mtcars |> mutate(wt/hp)`
- `with(mtcars, wt/hp)`

The building blocks of code evaluation

There are four building blocks used in evaluating code.

- **Constants**: A specific value like `1` or `"data/study.csv"`.
- **Symbols**: A name of an object, like `pi`.
- **Expressions**: Code structured as an AST.
- **Environments**: The place where named objects are found.

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Question?

How are these building blocks used together to construct and evaluate code?

The building blocks of code evaluation

In `rlang`, we have three main building block functions:

- `sym("pi")`: a symbol/name like `pi`
- `expr(1/pi)`: an expression for `1/pi`
- `quo(1/pi)`: a *quosure* (expression **and** environment)

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Follow along!

Use `call2()` and these building blocks to construct and evaluate `mtcars |> mutate(wt/hp)`.
Hint: recall that `x |> f(y)` is parsed as `f(x, y)`.

The building blocks of code evaluation

Your turn!

Spot the difference.

How do the results of the following functions differ?

- `sym("2 * pi")`
- `expr(2 * pi)`
- `quo(2 * pi)`

Capturing code

More often than not, NSE involves capturing user code that was used in your function. This is done with `en*()` functions:

- `ensym(x)`: capture a symbol
- `enexpr(x)`: capture an expression
- `enquo(x)`: capture a quosure

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- `enquo(x)`: capture a quosure

These must be used inside functions, for example:

```
capture_expr <- function(x) {
  enexpr(x)
}
capture_expr(1/pi)
```

Unquoting (bang-bang!!)

Why doesn't the following code work?

```
log_expr <- function(x) {
  # Capture expression
  x <- enexpr(x)
  # Return new expression with log()
  expr(log(x))
}
log_expr(1/pi)
```

```
log(x)
```

Unquoting (bang-bang!!)

To use captured code in our functions, we need to unquote it.

```
log_expr <- function(x) {
  # Capture expression
  x <- enexpr(x)
  # Return new expression with log()
  expr(log(!!x))
}
log_expr(1/pi)
```

```
log(1/pi)
```

`expr(log(!!x))` will create an expression (`expr()`) that replaces `x` with its value (`1/pi`).

Unquoting (bang-bang!!)

Unquoting in analysis

Unquoting replaces the object's name with its value.
This is also useful when using NSE functions.

How can `!!` be useful with dplyr?

Unquoting (bang-bang!!)

Suppose we wanted to programmatically `filter()` `mtcars$cyl`:

```
cyl <- 4
mtcars |>
  filter(cyl == cyl)
```

What's the problem? How can unquoting help?

Embracing inputs ({{curly-curly}})

The pattern `!!enquo(x)` is so often in functions that it has a special shortcut known as 'embrace' or 'curly-curly'. The code `{x}` is identical to `!!enquo(x)`.

Consider this function for summarising a value's range:

```
var_summary <- function(data, var) {
  data |>
    summarise(n = n(), min = min({{ var }}), max = max({{ var }}))
}
mtcars |>
  group_by(cyl) |>
  var_summary(mpg)
```

Why is `enquo()` important here?

Unquote-splicing (bang-bang-bang!!!)

It is sometimes useful to unquote multiple code elements across multiple arguments of a function.

This is done with unquote-splicing using `!!!` on a list of symbols, expressions, or quosures.

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A list symbols, expressions, or quosures can be:

- created with `syms()`, `exprs()`, `quos()`
- captured with `ensyms()`, `enexprs()`, `enquos()`

This is often used to capture, modify and pass on dots (...).

Unquote-splicing (bang-bang-bang!!!)

For example, the `var_summary()` function can be extended to accept multiple variables (or expressions) via dots (. . .).

```
var_summaries <- function(data, ...) {
  vars <- enquos(...)
  .min <- purrr::map(vars, ~ expr(min(!!.)))
  .max <- purrr::map(vars, ~ expr(max(!!.)))
  data |>
    summarise(n = n(), !!!.min, !!!.max)
}
mtcars |>
  group_by(cyl) |>
  var_summaries(mpg, wt)
```

Tidy dots (:=)

Tidy dots (`:=`) allow the argument names to be unquoted too.

For example:

```
my_df <- function(x) {
  tibble(!!expr_text(enexpr(x)) := x * 2)
}
my_var <- 10
my_df(my_var)
```

```
# A tibble: 1 x 1
  my_var
   <dbl>
1     20
```

You can alternatively use `!!!` with a named list.

Tidy evaluation

Tidy evaluation refers to the use of NSE in the tidyverse to make data analysis easier.

NSE is used widely across tidyverse packages, but at the same time it is used sparingly.

Your turn!

Question
Where have you seen NSE used in tidyverse packages?

Tidy evaluation

Tidy evaluation searches the variables of the data first, followed by the search path of the user's environment.

This is a type of NSE, since it changes the environment in which code is ran.

```
mtcars |>
  mutate(mpg/wt)
```

`mpg/wt` would ordinarily error since `mpg` and `wt` aren't found, but `mutate()` uses NSE to first search the dataset.

Tidy evaluation

This is accomplished using `eval_tidy()`, with the arguments:

- `expr`: The expression (code) to evaluate
- `data`: The dataset 'mask' to search first
- `env`: The environment to search next.

Unlike `eval()`, this will:

- Respect the environments of quosures
- Attach *pronouns* for `.data` and `.env`

Tidy evaluation

We can use `eval_tidy()` to create a simple `dplyr::mutate()` function variant.

```
my_mutate <- function(.data, mutation) {
  mutation <- enquo(mutation)
  result <- eval_tidy(mutation, data = .data, env = caller_env())
  .data[[as_label(mutation)]] <- result
  .data
}
mtcars |>
  my_mutate(mpg/wt)
```

Question: What features are missing in our function compared to `dplyr::mutate()`?

tidyselect

The tidyselect package is useful for selecting variables from a dataset with NSE.

You almost certainly have used it in the tidyverse without knowing.

It powers column selection in:

- `dplyr` for `select()`, `across()`, and more.
- `tidyr` for almost everything.

tidyselect

It enables variable selection with a domain specific language (DSL), which uses NSE to identify columns with:

- `var1:var10`
- `matches("x.\\d")`
- `all_of(<chr>)`
- `where(<fn>)`

tidyselect

If you need tidy column selection, simply import and use `tidyselect::eval_select()`.

```r
library(tidyselect)
x <- expr(mpg:cyl)
eval_select(x, mtcars)
```

```
mpg cyl
  1   2
```

This function returns the column numbers that were selected.

tidyselect

Putting it all together, we can create our own `dplyr::select()` function variant.

```
my_select <- function(.data, cols) {
  cols <- eval_select(enexpr(cols), .data)
  .data[cols]
}
my_select(mtcars, c(mpg, wt, vs:carb))
```

Your turn!

Modify this function to instead accept the selected columns via the dots (...), just like `dplyr::select()` does.

Software design

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But once again, with great power…

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But once again, with great power…

… comes great responsibility!

Tidyverse design principles

Notice how little NSE the tidyverse uses to great effect.

A lot of thought has gone into designing the tidyverse, which mostly uses standard evaluation: https://design.tidyverse.org/

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A design compromise

While very appreciated by users, NSE introduces a lot of complexity when programming with tidyverse packages.

Software design with NSE

In most cases you shouldn't add NSE to your package.

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Understanding NSE however is very useful for advanced use of tidyverse packages in non-interactive contexts.

Software design with NSE

If you must use NSE, you should:

- Use it sparingly
- Be consistent
- Clearly document it
- Get a lot of design benefit from it

  (not just for slightly less typing!)